Formalizing the IUT III 3.11–3.12 Corridor in Lean

IUT Lean formalization project

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Abstract

This paper reports the current state of a Lean formalization project for a narrow part of Mochizuki's inter-universal Teichmuller theory: the corridor from IUT III, Theorem 3.11 to Corollary 3.12. The project is deliberately neutral about the correctness of IUT as a proof of the *abc* conjecture and about the Mochizuki–Scholze–Stix dispute. Its purpose is to make the comparison levels, indeterminacy actions, real-line transports, and source obligations explicit enough that the proof assistant can distinguish proved consequences from mathematical hypotheses still to be constructed.

The formalization now verifies the ordered-real endpoint of Corollary 3.12, separates the representative j^2-level from full-label, averaged, and hull/log-volume comparison levels, and builds a typed route through concrete Hodge-theater/log-theta-style source data, $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$indeterminacy records, quotient possible images, a selected q-region, Remark 3.9.5-style hull/determinant data, and the existing C_Θ endpoint. The current result is a checked conditional Stage 1 corridor, not a closed proof of Corollary 3.12 from Mochizuki's published hypotheses.

We also maintain a Lean blueprint and intend to make the project consumable by Apodeixis as a collaboration and review layer. The blueprint is presently a compact navigational document plus a generated declaration index; it is useful for orientation, but it is not yet a complete mathematical dependency graph. This paper replaces an earlier ledger-style draft. Future updates should revise the relevant sections of this research paper rather than append chronological implementation notes.

1 Introduction

IUT is large enough that a useful formalization cannot begin by translating all four papers in order. The present project instead focuses on the public pressure point around IUT III, Theorem 3.11 and Corollary 3.12. Mochizuki's 2026 formalization report identifies this as the first stage of a possible formalization program:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The label “3.11.5” is not a theorem in the IUT III paper. It names the reorganized checkpoint obtained by moving the holomorphic hull plus determinant operation before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The immediate formal target is not the *abc* conjecture. It is the logical shape of the Stage 1 implication. The question is:

> If the source constructions claimed by IUT III supply the multiradial possible images, indeterminacy behavior, IPL, SHE, APT, hull/determinant operations, and local-to-global C_Θ estimates, does the Corollary 3.12 inequality follow in a type-correct way?

The current code answers a substantial conditional version of this question. It proves the final ordered-real implication, constructs many intermediate typed routes, and exposes the remaining source mathematics as named obligations. It does not yet prove that the full Hodge-theater, Frobenioid, log-Kummer, holomorphic hull, determinant, and IUT IV analytic estimates of the source papers instantiate those obligations.

2 Mathematical Target

2.1 The final inequality

The endpoint formalized in Lean is the signed real comparison used in the Corollary 3.12 corridor. In the notation of the project, the route must supply

$$q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}}, \qquad 0 < -q_{\mathrm{signed}}, \qquad \Theta_{\mathrm{signed}} \le C_\Theta(-q_{\mathrm{signed}}).$$

From these hypotheses the ordered-real calculation gives

$$-1 \le C_\Theta.$$

The Lean theorem behind this step is elementary. This is important: the hard question is not whether the final real inequality follows from the displayed signed comparisons. The hard question is whether the source machinery really constructs the signed comparison without collapsing distinct histories or comparison levels.

2.2 The disputed comparison level

Scholze and Stix criticize the proof around Corollary 3.12 on the grounds that the relevant Θ-side data carries j^2-scaled information, while a naive simplification of the comparison appears to remove the room needed for that data. The formalization responds by enforcing separate types for:

representative j^2-coordinate data,
balanced sign-compatible data,
full-label and averaged finite-label data,
transported ordered-real copies,
hull/log-volume endpoint data.

Lean then rejects the most naive collapse: the balanced sign-compatible layer does not by itself supply the final Corollary 3.12 route. This is a diagnostic result, not a settlement of the dispute. The remaining issue is whether the source construction gives the final hull/log-volume object with the required compatibilities.

2.3 Indeterminacy behavior

The current Stage 1 interface treats the three indeterminacy components separately. In the formal model, Ind_1 and Ind_2 preserve procession-normalized log-volume and generate equality for the possible-image quotient. Ind_3 is not an equality generator; it contributes only an upper-semi log-volume inequality. This distinction is now part of the checked Lean interface:

$$\mathrm{Ind}_1, \mathrm{Ind}_2 : \text{equality/invariance}, \qquad \mathrm{Ind}_3 : \text{upper-semi inequality}.$$

This separation is one of the central design decisions of the formalization. It prevents a proof from turning an upper-semi source relation into an equality merely because both appear in the informal phrase “indeterminacy”.

3 Formalization Architecture

3.1 Source, Lean, paper

The working cycle is:

$$\text{Mochizuki source papers} \longrightarrow \text{Lean definitions and theorems} \longrightarrow \text{research paper and blueprint.}$$

The papers in `docs/` are treated as source of truth. The Lean code is the formal artifact. This paper records the mathematical state of the formal artifact; it is not meant to be a chronological log. This distinction matters because previous versions of this draft had grown into an implementation ledger. The present version is organized as a research paper: target, architecture, results, trust boundary, tooling, and next work.

3.2 Main Lean layers

The Stage 1 development is split into modules whose roles are stable enough to describe at paper level:

Module family	Role
`PilotComparison`, `CorollarySchema`, `SourceObligations`	The signed real endpoint, the final Corollary 3.12 predicate shape, and the ledger turning a common-bound comparison into the endpoint.
`IUTStage1SourceCore`, `IUTStage1FiniteLabels`, `IUTStage1Gaussian`	Real-line copies, regions, finite-label and $ZMod(\ell)$ models, Gaussian degree shadows, and j^2-diagnostic infrastructure.
`IUTStage1StepX`, `IUTStage1HodgeSHE`	Procession-normalized log-volume routes, Hodge/SHE data, square/full-label transport, and upper-semi compatibility records.
`IUTStage1Theorem311`, `IUTStage1StepXI`	Typed Theorem 3.11 indeterminacy cores, possible-image quotient structures, concrete Hodge-theater/log-theta source layers, Remark 3.9.5 hull/determinant interfaces, and paper-trace audits.
`IUTStage1FrobenioidShift`, `IUTStage1Experiments`	Nonarchimedean realified Frobenioid/log-Kummer and vertical-IQ routes, additive-Haar arithmetic-degree endpoints, and compiled public diagnostics.

The names in this table are intentionally coarse. A paper should not reproduce thousands of declaration names. The detailed declaration surface is maintained in the blueprint's generated declaration index.

3.3 Records as boundaries

The project uses records in two different ways. Some records are ordinary mathematical data structures whose fields are proved or constructed in Lean. Other records are trust-boundary interfaces: they name a source construction that must eventually be derived from the IUT papers. The current paper distinguishes these roles explicitly.

For example, the typed indeterminacy core is a useful formal object: it stores the $\mathrm{Ind}_1/\mathrm{Ind}_2$ invariance laws, the Ind_3 upper-semi law, and the quotient relation. By contrast, the full Theorem 3.11-and-remarks obligation record still contains source-facing fields such as construction of the input prime-strip link, theta-pilot possible images, the log-theta-lattice procession, and the Hodge/SHE/IPL/APT bridge. These fields are not hidden axioms, but they are also not yet source-derived mathematics.

4 Current Formal Results

4.1 Ordered-real endpoint

The final ordered-real implication is formalized. Once the signed comparison $q_{\text{signed}} \leq \Theta_{\text{signed}}$, positivity of $-q_{\text{signed}}$, and the C_Θ-upper bound are available, Lean derives the expected lower bound $-1 \leq C_\Theta$. The code also proves the corresponding strict/boundary dichotomy: either the boundary case is exact, or $-1 < C_\Theta$.

This part is no longer an open mathematical problem inside the project. It is pure ordered-real algebra. All serious mathematical work sits upstream of the signed comparison and the C_Θ-upper bound.

4.2 Anti-collapse diagnostics

The formalization proves negative diagnostic statements about comparison-level collapse. In particular, a label-independent j^2 collapse is rejected at the representative level, and the balanced sign-compatible layer is not accepted as the final route level. These diagnostics are valuable because they test the exact failure mode emphasized in external criticism.

The diagnostics should be read carefully. They show that the Lean development does not silently make the criticized simplification. They do not show that Mochizuki's construction successfully supplies the stronger hull/log-volume data.

4.3 Concrete Stage 1 route

The current strongest route starts with a unified Hodge-theater/log-theta/log- Kummer source spine. This spine carries the 0- and 1-column histories, prime-strip data, theta-pilot possible images, the selected q-choice, and the vertical log-Kummer packet alignment. Its Theorem 3.11 bridge side is also structured: the input-prime-strip link, one-column simultaneous holomorphic expression, and algorithmic parallel transport are projections of one IPL/SHE/APT bridge package. The Step (x) side is represented by a single finite-divisor/vertical-log-Kummer package: finite divisor tensor packets, realified Frobenioid Kummer compatibility, mono-analytic forgetting, vertical IQ, and source-target log-volume calibration are projections of one object rather than five unrelated source flags. Lean projects from the same spine both the Theorem 3.11 typed $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$ core and the Step (x) payload. The route then forms the equality-quotient family of possible images, selects the q-region, and connects that region to the existing C_Θ/Corollary 3.12 corridor. The preferred paper-trace route now has a named source-spine audit: the Theorem 3.11/Remarks obligation package and the Step (x) finite-divisor/log-Kummer obligation package are projected from one source object. The first Step (xi) hard-math route has also been added: a paper-derived Remark 3.9.5/Step (xi) source records the substeps (xi-a)–(xi-g), ties the selected q-region to the actual Theorem 3.11 possible image selected by the source spine, and projects the existing Step (xi) hull/determinant obligation record internally. Ob7 is no longer only a raw compatibility flag on the newest route: the legacy prime-strip/log-Kummer record and its input/output strip and selected-column equalities are projected from a coric $\Theta^{\times\mu}$/log-Kummer source link aligned with the same quotient hull source. The selected q-region containment is now derived from the equality-quotient possible-image family: it is the canonical inclusion of the selected possible region into the family union, followed by the constructed holomorphic hull source. The paper-derived Step (xi) source also has a quotient-compatible constructor that fixes the selected choice to the source spine, instead of accepting an independently assembled hull record at that point. The remaining IUT IV and additive-Haar layers are then combined with this constructed Step (xi) source. A goal-facing completion audit packages this milestone into checked claims:

paper trace, concrete choice space, typed indeterminacy core,
procession/F_ℓ, quotient possible images, selected q-region,
Remark 3.9.5 bridge, constructor into the corridor,
lowered endpoint, audit/paper handoff.

The audit is a projection of the assembled remaining-payload record. It does not add a new mathematical assumption. Its function is to make the project milestone checkable and reviewable.

4.4 Local-to-global C_Θ chain

The code now contains a proof-carrying local-to-global C_Θ route at the interface level. The preferred additive-Haar arithmetic-degree and p-adic formula route threads local analytic estimates, arithmetic-degree calibration, finite-place Haar defects, formula-gap matching, and Step (xi) local terms into the endpoint. For this preferred route, the canonical C_Θ-scale comparison is derived from the finite-place arithmetic gap and the IUT IV handoff identity, rather than supplied as a raw endpoint hypothesis.

The phrase "proof-carrying" is deliberate. The route no longer consumes a single unstructured numerical inequality at the public endpoint. It consumes a structured chain of local and global source objects. Nevertheless, several of those source objects still stand for arithmetic constructions not yet derived from number-field valuations, conductors, differents, holomorphic hulls, and the local estimates of IUT IV.

5 What Remains Mathematically Open

5.1 Theorem 3.11 from source data

The largest remaining mathematical task is to construct the Theorem 3.11 multiradial output from actual IUT I–III source data. The present code has a unified Hodge-theater/log-theta/log-Kummer source spine that projects to the typed core, possible-image quotient, input-prime-strip link, SHE/IPL/APT bridge, and Step (x) finite-divisor/log-Kummer payload. The preferred paper-trace API no longer asks for those Theorem 3.11 and Step (x) pieces as parallel flags; they are read from the spine. The implementation does not yet derive that spine, or the structured bridge and Step (x) packages inside it, from the full Hodge-theater, local LGP, and Frobenioid constructions.

The missing source construction includes:

> initial theta data and Hodge-theater histories,
> prime strips and the derivation of the structured input link,
> the multiradial algorithm and its possible images,
> the log-theta-lattice procession and F_ℓ-symmetry,
> the derivation of the 1-column SHE/IPL/APT bridge,
> the derivation of the Step (x) package from vertical log-Kummer data,
> the proof that $\mathrm{Ind}_1/\mathrm{Ind}_2$ preserve normalized log-volume,
> the proof that Ind_3 has only the stated upper-semi orientation.

The Lean records now say exactly what must be supplied. They do not yet replace Mochizuki's proof of Theorem 3.11.

5.2 Remark 3.9.5 and Step (xi)

The second major gap is the genuine holomorphic hull and determinant construction. The current code now has a paper-traced Step (xi) source route: the old Step (xi) obligation record can be projected from a structured source that records Corollary 3.12 Step (xi-a)–(xi-g), Remark 3.9.5 Ob1–Ob9, the selected Theorem 3.11 possible image, determinant data, and Ob7 link pinning. At the hull-formation boundary, the selected q-region is now constructed from the equality-quotient possible-image family, so its containment in the possible-image union is no longer a separately supplied Step (xi) field; the paper-derived source constructor uses the source spine's selected q-choice directly. At the determinant boundary, the Step (xi) determinant comparison record can now be constructed from the weighted arithmetic-vector-bundle determinant source already present in the Step (xi) layer: the finite adjusted summands, determinant sum, normalization, and positive tensor power are projected from that lower source rather than supplied as independent fields. The record-canonical Remark 3.9.5 hull/determinant bridge now also feeds this record: once the paper-derived selected q-region log-volume is identified with the record bridge q-pilot log-volume, the q-region-to-normalized- determinant bound and the determinant-to-Θ_{signed} bound are both projected from the bridge. The bridge-backed paper-source constructor now derives this log-volume identification from lower alignment data: equality of the paper hull operator with the bridge hull operator, and equality of the bridge q-pilot region with the source-spine selected Theorem 3.11 possible image. The constructed-source version projects the bridge and hull operator from the constructed Remark 3.9.5 source itself, leaving only the q-pilot region/source-spine selected possible-image identification at this boundary. A further source-selected constructor forms the constructed Remark 3.9.5 source with the source-spine selected possible image as its q-pilot region, so this identification is definitional on that route; the remaining explicit condition is containment of the selected possible image in the record possible-image union. The Ob7 prime-strip/log-Kummer record can now be read from a single coric $\Theta^{\times\mu}$-style source link aligned with the same quotient-hull bridge; its retained statement includes the local Gaussian/environment equalities and unit-character equalities from the Ob7 source endpoint. The strongest selected route now constructs this source link from the constructed Remark 3.9.5 bridge and a coric prime-strip invariant, after transporting determinant compatibility across the record-union/equality-quotient-family alignment. In the same-index source-spine case, this alignment is now derived from the equality of the record possible-image family with the Theorem 3.11 source image family; the selected route also derives the selected-q containment in the record union from the same equality. A lower selected route now takes the paper-shaped pointwise trace that the record possible image attached to each concrete Hodge-theater/log-theta choice equals the corresponding source-spine possible image; Lean assembles this trace into the full family equality before applying the quotient-union theorem. This pointwise trace is now itself derived from the existing theta-pilot class-region source once its class formula is identified with the Hodge-theater source spine's theta-pilot class-image family. At the public paper-trace route boundary, the remaining payload audit has now been split so the caller supplies only the non-Step-(xi) payload; the legacy Step-(xi) audit fields are derived internally from the structured Step-(xi) source. This is progress below a free-floating obligation bundle. What remains is to construct, from the source papers, the actual lower operations corresponding to:

the concrete holomorphic hull operator/source,
the lower weighted determinant/localization source itself,
the lower source proof of Ob1/Ob2 hull absorption,
Ob3/Ob4 adjusted determinant data from localizations,
Ob5 quotient determinant compatibility,
the lower identification of the class-region formula with source-spine class images,
the lower construction of the coric Ob7 prime-strip invariant.

This is the next natural place to eliminate a family of `Prop`-level obligations and replace it by constructed Lean data.

5.3 IUT IV estimates

The C_Θ comparison depends on IUT IV local and global estimates. The current formalization organizes the route through additive-Haar arithmetic-degree and p-adic formula objects, but the deepest estimates are still source obligations. What is now checked for the preferred route is the ordered-real and finite-sum handoff from the arithmetic gap to the canonical C_Θ-scale comparison. What remains to construct from first principles is the lower IUT IV source of that arithmetic gap:

the distinguished log-shell inclusions and numerical bounds,
the nondistinguished zero log-volume contribution,
the archimedean metric containment,
the arithmetic divisor source for Theorem 1.10,
the distinguished and archimedean formula-to-gap bounds,
the finite-place summation comparison.

Until these estimates and the upstream source spine are constructed from the published source data, the corridor remains conditional on source-paper mathematical content.

5.4 Local arithmetic and p-adic analytic infrastructure

The implementation has moved below a purely formal real-number boundary, but it still lacks significant local arithmetic infrastructure. The remaining work includes finite extensions of $\mathbb{Q}_p$, residue fields, Haar-normalized unit balls, explicit local degrees, different and conductor terms, and the formula matching that connects those objects to the arithmetic-degree route. This is standard mathematical infrastructure in spirit, but it is not small work in Lean.

6 Blueprint and Apodeixis

6.1 Lean blueprint status

The repository contains a Lean blueprint under `blueprint/`. It has a compact prose structure: aim and scope, the source Stage 1 route, the formal interface, the route to Corollary 3.12, open source obligations, and later work. It also has a generated declaration index, `blueprint/lean_decls`, which currently lists thousands of Lean declarations.

This means the blueprint is not empty, but it is still intentionally slim. It is adequate as a navigational document for the present stage. It is not yet a complete dependency graph of the project in the strongest sense: many nodes are marked not ready, and the prose chapters do not yet expand every major Lean route into a polished theorem-by-theorem mathematical dependency tree.

The right standard for the blueprint is different from the standard for this paper. The blueprint should be a map: concise nodes, Lean links, readiness status, and dependencies. This paper should be an argument: what we are formalizing, why the type boundaries matter, what is proved, and what remains mathematically open.

6.2 Apodeixis integration plan

The project is intended to be compatible with Apodeixis as a collaboration and review layer. The practical goal is to expose the formalization in a way that supports external mathematical review:

$$\begin{gathered}\text{source-paper claim}\\ \Downarrow\\ \text{blueprint node}\\ \Downarrow\\ \text{Lean declaration or source obligation}\\ \Downarrow\\ \text{audit status and review discussion.}\end{gathered}$$

For this to work, the Lean code must avoid opaque omnibus assumptions, the blueprint must keep source claims linked to declarations, and the paper must avoid becoming a changelog. The current audit records are designed with this future use in mind. They give Apodeixis a natural unit of review: a named mathematical obligation with a precise Lean consumer.

7 Methodological Lessons

7.1 Typed boundaries are productive

The main technical lesson so far is that typed boundaries are not bureaucracy. They are mathematical tests. Once ordered real-line copies, transports, finite-label quotients, hull regions, source certificates, and upper-semi relations are assigned different types, informal shortcuts become impossible unless they are backed by explicit constructors.

This is especially useful in the 3.11–3.12 corridor because the public dispute is about whether a comparison has lost information during transport between different mathematical histories. Lean cannot decide that philosophical issue by itself, but it can prevent a formal proof from hiding the relevant movement.

7.2 Conditional routes are still useful

A conditional theorem can be scientifically useful if it names its conditions precisely. The current Stage 1 route is not a proof of Corollary 3.12 from the published IUT papers, but it has lowered the ambiguity of the problem. The remaining obligations are no longer "somewhere in the comparison". They are organized into Theorem 3.11 source construction, Step (xi) hull/determinant construction, IUT IV local-to-global estimates, and local arithmetic infrastructure.

7.3 The paper must not be a ledger

The earlier draft recorded too many implementation increments. That made it long but not more informative. The standing policy for future updates should be:

update the relevant section,
replace stale claims,
move only stable mathematical outcomes into the paper,
leave declaration-level detail to Lean and the blueprint.

This paper should remain regular research-paper length. The blueprint and generated declaration index are the correct place for detailed navigation.

8 Roadmap

The next milestones should eliminate source-obligation families rather than adding more endpoint wrappers.

Milestone A: Step (xi) hull/determinant construction. Replace the current Remark 3.9.5/Step (xi) obligation fields by constructed holomorphic hull and determinant data. This is the most direct way to advance the 3.11–3.12 issue, because it targets the passage from multiradial possible images to the signed q-region comparison.

Milestone B: IUT IV local-to-global C_Θ. Construct the local estimates and finite-place summation chain underlying the canonical C_Θ-scale comparison. This would remove the deepest remaining numerical assumption in the current endpoint.

Milestone C: Theorem 3.11 source derivation. Work backward from the typed indeterminacy core to actual Hodge-theater, Frobenioid, log-Kummer, and log-theta-lattice source data. This is the largest milestone and the one most directly connected to the Scholze–Stix objection.

Milestone D: Blueprint expansion. Expand the blueprint from a slim guide plus declaration index into a richer review graph. Each nontrivial source obligation should have a node with source citations, Lean declarations, readiness status, and a short explanation of the mathematical construction still missing.

9 Conclusion

The project has reached a useful but conditional Stage 1 state. Lean verifies the final real inequality, distinguishes the disputed comparison levels, keeps $\mathrm{Ind}_1/\mathrm{Ind}_2$ equality behavior separate from Ind_3 upper-semi behavior, and threads a concrete Hodge-theater/log-theta-style source route into the current C_Θ/Corollary 3.12 endpoint. The code is not merely a prose annotation of the papers; it is an executable audit of a precise corridor.

What remains is real mathematics, not formatting. To settle the 3.11–3.12 issue inside this project, the source obligations must be discharged from Mochizuki's constructions: full Theorem 3.11 multiradial data, genuine Remark 3.9.5 hull/determinant operations, IUT IV local estimates, and the local arithmetic infrastructure connecting them. The formalization has made that task sharper. It has not made it disappear.

References

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